

In particular, this implies that if $p \neq 2, 3$ and $p \in \pi(m)$, then

$$(1.3) \quad p \geq \begin{cases} (m-1)^{\frac{1}{6}} & m \text{ is a prime,} \\ m^{\frac{1}{12}} & m \text{ is squarefree,} \\ m^{\frac{1}{24}} & \text{otherwise.} \end{cases}$$

Let $j(E_i)$ be the j -invariant of E_i , which is an algebraic integer, and $\varphi_m \in \mathbb{Z}[X, Y]$ the m -th modular polynomial. Then we can characterize the set $\pi(m)$ as

$$(1.4) \quad \pi(m) = \{p \text{ prime} : p \mid \text{Nm}(\varphi_m(j(E_1), j(E_2))), p \nmid m\}.$$

The following result is a direct consequence of Theorem 1.1 and offers an S -unit generalization of the main result in [Li21].

Corollary 1.6. *Suppose D_1, D_2 are fundamental discriminants and co-prime. For any finite set S of primes, the algebraic integer $\varphi_m(j(E_1), j(E_2))$ is not an S -unit for all $m \gg_\epsilon (\#S)^{5+\epsilon}$.*

We now briefly outline the proof of Theorem 1.1. The proof proceeds along a similar line of argument in [Li21], utilizing higher Green functions and theorems of Gross-Zagier and Gross-Kohnen-Zagier. The main innovation is to control the valuation of $\text{Nm}(\varphi_m(j(E_1), j(E_2)))$ at p by relating it to Fourier coefficients of cusp forms with increasing level. This is possible as such valuations are Fourier coefficients of incoherent Eisenstein series with full level, and can be approximated by Fourier coefficients of coherent Eisenstein series with higher levels, which are holomorphic Hilbert modular forms. This is the content of Theorem 2.5. After this, we still need a bound of such Fourier coefficients in terms of the level. This is achieved in Theorem 3.5 using Deligne's bound and a bound of the Petersson norm of a cusp form in terms of its Fourier coefficients at the cusp ∞ . Section 4 contains the proof of Theorem 1.1.

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2. PRELIMINARY

We follow [Li21] to recall some preliminary notions.